

From flower to fruit



WINTER FEAST
Fruits are an important source of winter food for many animals. Here, a redwing feasts on fallen apples and may help to spread the apples' seeds.

PLANTS GROW FLOWERS for an all-important task: to make seeds and spread them to new places. Some flowers make just one seed, while others can make millions. But no matter how many they make, the steps they follow are usually the same. For most flowers, the first step is pollination (pp. 22–23). This transfers tiny male pollen grains from flower to flower. If a pollen grain lands on the right kind of flower, step two can begin. The pollen grain comes to life and grows a microscopic tube down through the female part of the flower. Here, it searches out a female cell, or ovule. Once the pollen grain has found an ovule, its male cells travel down the tube and into the ovule to fertilize it. The pollen grain has now done its work, and the fertilized ovule starts to turn into a seed. Meanwhile, step three gets under way. The flower—now without its petals—begins to turn into a fruit.

A fruit is something that helps the seeds to spread. Some plants produce soft, juicy fruits that attract animals looking for a meal. Animals eat the fruit, and they scatter the seeds as they feed. Many more plants produce dry, hard fruits that break open when they are ripe. These fruits are not designed to be eaten. Rather, the seeds inside are spread by other means (pp. 28–29).

EARLY DAYS

Even before the rose comes into flower, the beginnings of the rose hip can be seen clearly. The top of the stem, to which the flower parts are attached, is swollen and globular, as this bud shows. The flower's female parts, the ovaries and ovules, are inside this swollen area, which is called a receptacle.

Green sepals fold back

THE ROSE IN BLOOM

As the bud opens, the flower gives off its sweet scent, which attracts bees for pollination (pp. 22–23). Once the flower has been pollinated and the ovules fertilized, the receptacle begins to swell.

Sepals protect the developing bud

Receptacle containing ovaries

Petals open out

Thorns keep leaf-eating animals away